

SAURABH TANDON

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I'm an M.Tech scholar in AI & Machine Learning at BITS Pilani and an AI Engineer with over **4.5 years** of experience focused on building advanced LLM-powered systems. I design intelligent multi-agent scalable workflows using LangGraph and LangChain with a strong foundation in ML, NLP, Deep Learning, statistics, and SQL, I create systems that go beyond data processing-enabling reasoning, adaptability, and real business impact.

WORK EXPERIENCE

Blueflame Labs Pvt. Ltd.

03/2025 - Present

AI Specialist

- Architected a **production-grade, event-driven multi-agent HRMS automation platform** leveraging **LangGraph (StateGraph, conditional edges, human-in-the-loop checkpoints, persistent memory)** and **LangChain tool-binding with dynamic routing**, handling 10+ enterprise workflows with **scalability** enabling zero-lag concurrent user sessions.
- Designed and deployed a **hierarchical chunking + parent-child retrieval RAG pipeline** using **LangChain RecursiveCharacterTextSplitter, multi-vector retriever, and cross-encoder reranking (BGE-reranker)** over an **Azure AI Search + FAISS hybrid vector store**, achieving high-precision grounded responses on internal HR corpora.
- Engineered an **MCP (Model Context Protocol) server** with real-time REST/SSE transport exposing live internal job openings feed, enabling employees to refer.
- Implemented **AES-256 encrypted at-rest storage** for employee PII within vector indices using **LangChain custom document transformers**; enforced **RBAC at retrieval layer**; maintained tamper-evident audit logs via **Azure Monitor + structured Python logging** for SOC2-aligned compliance.
- Deployed via **Azure Bot Services + Azure App Service (containerized FastAPI backend)** embedded natively into **Teams Bot Framework SDK**, reducing HR operational overhead by ~50% and scaling to enterprise-wide self-service adoption.
- Built an **automated data cleaning pipeline on Databricks (PySpark, Delta Lake, Unity Catalog)** integrating **Pandas UDFs, sentence-transformer embedding models (all-MiniLM-L6-v2), and an LLM-based auto suggestion** for fixing the invalid fields in the dataset.

Capgemini

09/2024 – 03/2025

Generative AI Engineer

- Architected an **enterprise-grade semantic search and knowledge retrieval platform** using **LangChain Expression Language (LCEL), OpenAI GPT-4-turbo, FAISS HNSW indexing, and Streamlit**, enabling vector-similarity-based project discovery over 10K+ internal documents and cutting duplicate engineering effort by 25%.
- Built a **multi-format document ingestion and indexing pipeline** using **LangChain document loaders (PyPDF2, Unstructured), OpenAI text-embedding-ada-002, and chunking strategies (semantic + fixed-size overlap)** with metadata-enriched payloads for filtered retrieval.
- Implemented **LangChain ReAct Agents with custom tool definitions** to automate end-to-end project onboarding workflows, reducing 30+ hours/month of manual effort; monitored agent traces, token usage, and retrieval latency end-to-end via **LangSmith observability platform**.

- Designed an **end-to-end NLP sentiment intelligence pipeline** on structured customer feedback (surveys, CRM exports, reviews) using **Python (NLTK, spaCy, scikit-learn)** — encompassing tokenization, lemmatization, stopwords removal, and **TF-IDF + n-gram feature engineering** — training and ensembling **Logistic Regression, Random Forest, and Linear SVM classifiers** on 100K+ records persisted in **MySQL with optimized indexed schemas**.
- Applied **LDA topic modeling (Gensim)** with coherence score tuning to surface latent pain-point themes from unstructured feedback; delivered real-time insight dashboards via **FastAPI backend + Matplotlib/Plotly visualizations** on a custom **HTML/CSS frontend**, enabling data-driven product decisions.
- Established **reproducible ML workflows** with **scikit-learn Pipelines, joblib model serialization**, and versioned experiment tracking for iterative retraining cycles across quarterly feedback batches.

PROJECTS

Thyroid Detection

01/2023 – 03/2023

- Engineered **clinical-grade ensemble ML pipeline (Decision Tree, XGBoost, Random Forest)** on 12K+ records using **scikit-learn Pipelines, SMOTE-ENN resampling, RFE + SelectKBest feature selection, and GridSearchCV + StratifiedKFold** tuning; tracked experiments, AUC-ROC, and macro-F1 via **MLflow model registry**.
- Containerized inference microservice using **Docker multi-stage builds → AWS ECR → EC2 Auto Scaling** behind **Flask + Pydantic-validated REST API**; implemented **zero-downtime CI/CD via GitHub Actions** (lint → test → build → push → rolling deploy) with automated rollback triggers.
- Persisted patient records, feature vectors, and prediction logs in **MongoDB Atlas (indexed time-series collections)** with **audit trails and data versioning** across all inference events for full compliance-grade reproducibility.
- Used **Flask** to allow users to interact with the model through webpage. Technologies and libraries used are **Python, scikit-learn, MongoDB, Pandas, Docker, GitHub Actions, AWS**.

SKILLS

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|----------------|----------------|----------------|---------------|
| • Python | • Scikit-learn | • LLMs, OpenAI | • Bert, GPT |
| • NLTK | • LangChain | • CrewAI | • MLFlow, DVC |
| • MCP | • LangGraph | • Tensorflow | • Docker |
| • A2A | • LangSmith | • Keras | • Git Actions |
| • Hugging Face | • AutoML | • PyTorch | • AWS, Azure |

EDUCATION

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|---|-----------|
| BITS Pilani, WILP | 2024-2026 |
| • Course: M.Tech in Artificial Intelligence and Machine Learning , 7.77 CGPA | |
| PES Institute of Technology | 2017-2021 |
| • Course: B.E in Mechanical Engineering , 8.46 CGPA | |

CERTIFICATIONS and ACHIEVEMENTS

- Databricks Certified Generative AI Engineer Associate
- Azure Fundamentals AZ-900
- Deep Learning Specialization: DeepLearning.AI
- Won **Performer of the Year** award 2025-26